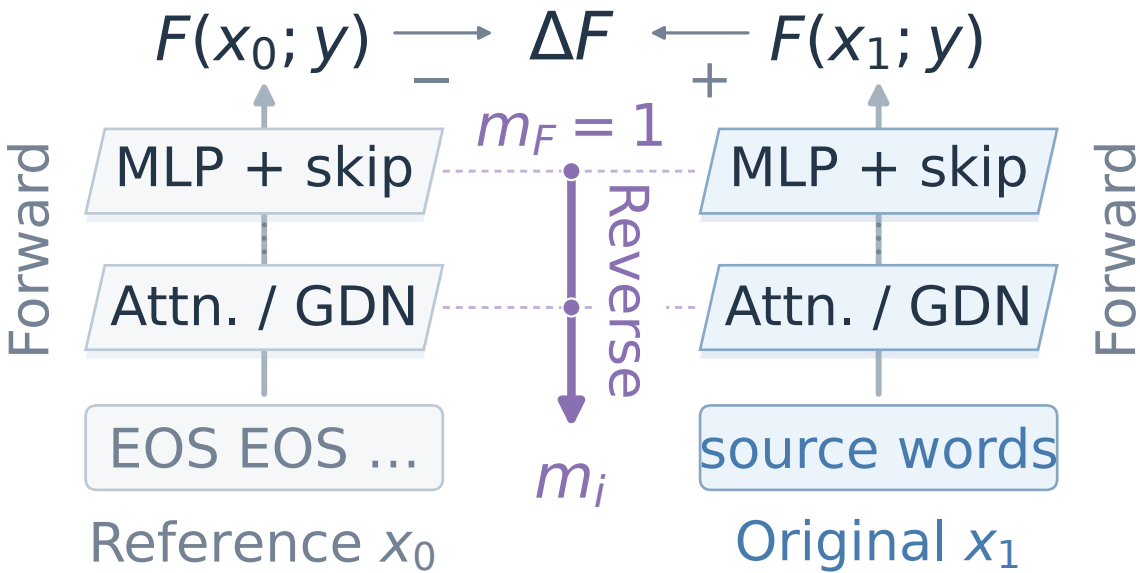


(a) From input change to credit

1 Two forward runs; fixed response y



2 Local rules

from both runs

$$\Delta h = D_f \Delta a$$

3 Reverse

propagate coefficients m

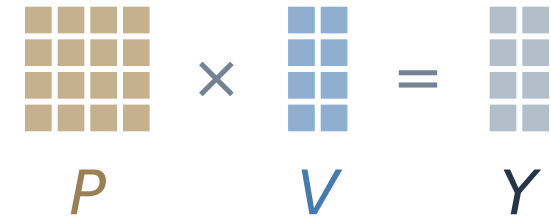
$$m_a = D_f^T m_h$$

4 Token score

$$A_i = \langle m_i, \Delta e_i \rangle$$

(b) Attention

Select and carry



0: reference; 1: original

Reverse

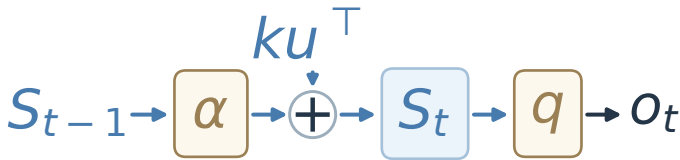
$$\Delta Y = \boxed{P_1 \Delta V} + \boxed{\Delta P V_0}$$

Change identity

$$M_V = P_1^T M_Y \quad M_P = M_Y V_0^T$$

(c) Gated DeltaNet

Retain, write, and read



Retention: $T = \alpha S$

Reverse

$$\Delta T = \boxed{\alpha_1 \Delta S} + \boxed{S_0 \Delta \alpha}$$

Change identity

$$M_S = \alpha_1 M_T \quad m_\alpha = \langle M_T, S_0 \rangle$$

(d) Choose the playwright, not the composer.

Qwen3.5 · MoreHopQA example 1

CONTEXT EXCERPTS

... William Shakespeare's play

Name-span total (nats)

+1.56

... The score is by William Walton.

-1.13

REQUESTED ROLE (EXCERPT) nats · color saturated

... the author of the play ...

≤ -2.5 0 ≥ +2.5

Shown spans underlined

Answer

William Shakespeare

$$\sum_i A_i = \Delta F$$

Across all input sources

(a) Qwen3-8B

Context

One of the special magic numbers for deafening-opium is: 5443951.

• • •

One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

Full input



(b) Qwen3.5-9B

Context

One of the special magic numbers for deafening-opium is: 5443951.

• • •

One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

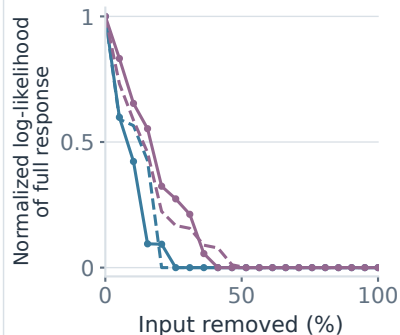
Full input



(c) Deletion

Answer

5443951, 8698256

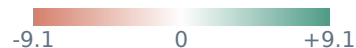


— Qwen3-8B DT

--- Qwen3-8B FT

— Qwen3.5-9B DT

--- Qwen3.5-9B FT

Signed contribution
(nats)

(a) Qwen3-8B

Context

Townsend Putnam Coleman III (born May 28, 1954) is an American voice actor

...
he also did additional voices in films "Fantasia 2000" (1999) and "Sing" (2016)
...

It was directed and written by Garth Jennings, co-directed by Christophe Lourdelet,

Question

How many consonants are there in the last name of the person who directed and wrote the 2016 film featuring the voice of Townsend Coleman?

Full input



(b) Qwen3.5-9B

Context

Townsend Putnam Coleman III (born May 28, 1954) is an American voice actor

...
he also did additional voices in films "Fantasia 2000" (1999) and "Sing" (2016)
...

It was directed and written by Garth Jennings, co-directed by Christophe Lourdelet,

Question

How many consonants are there in the last name of the person who directed and wrote the 2016 film featuring the voice of Townsend Coleman?

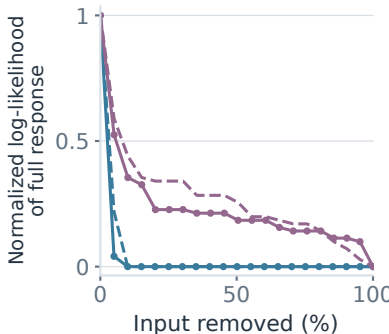
Full input



(c) Deletion

Answer

6 consonants in Jennings



- Qwen3-8B DT
- Qwen3-8B FT
- Qwen3.5-9B DT
- Qwen3.5-9B FT

Signed contribution (nats)

